

Perturbation evidence is multiplying faster than it can be organised

A single genome-scale Perturb-Seq study can profile **more than 2.5 million cells**, while pooled screens test thousands of targets. That scale could reveal robust biology across studies—but identifiers, metadata, assays and processing pipelines differ. Results remain hard to compare or reproduce. The goal of the Perturbation Catalogue is to supply a shared schema, stable gene identity and one access path for combining this evidence.

36× growth in perturbation papers over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: Replogle et al., Cell 2022.

Our approach turns scattered datasets into a unified resource

1,201 CRISPR screens

gene perturbation × phenotype or fitness

26 Perturb-Seq datasets

perturbation × single-cell transcriptome

10 MAVE datasets

variant × functional measurement

1,237 DATASETS

19,235 human targets

1,200 cell lines

36 tissues

169 diseases

34 cell types

We extensively curate metadata...

84 metadata fields

Study & provenance

Title, authors, year, source URI, licence, linked datasets

Perturbation & library

Targets, library design, delivery, guides, variants

Model system

Organism, tissue, cell type or line, disease, sex, stage

Readout & processing

Assay, technology, sequencing kit, reference, software

Stable Ensembl gene IDs and ontology-linked entities make the same biological question addressable across modalities.

And reprocess data from scratch

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.

SRA / FASTQ

Raw sequencing reads and study-specific probe lists

→

Cell × gene counts

Shared reference and gene annotation

→

Guide assignment

Map cells to perturbations

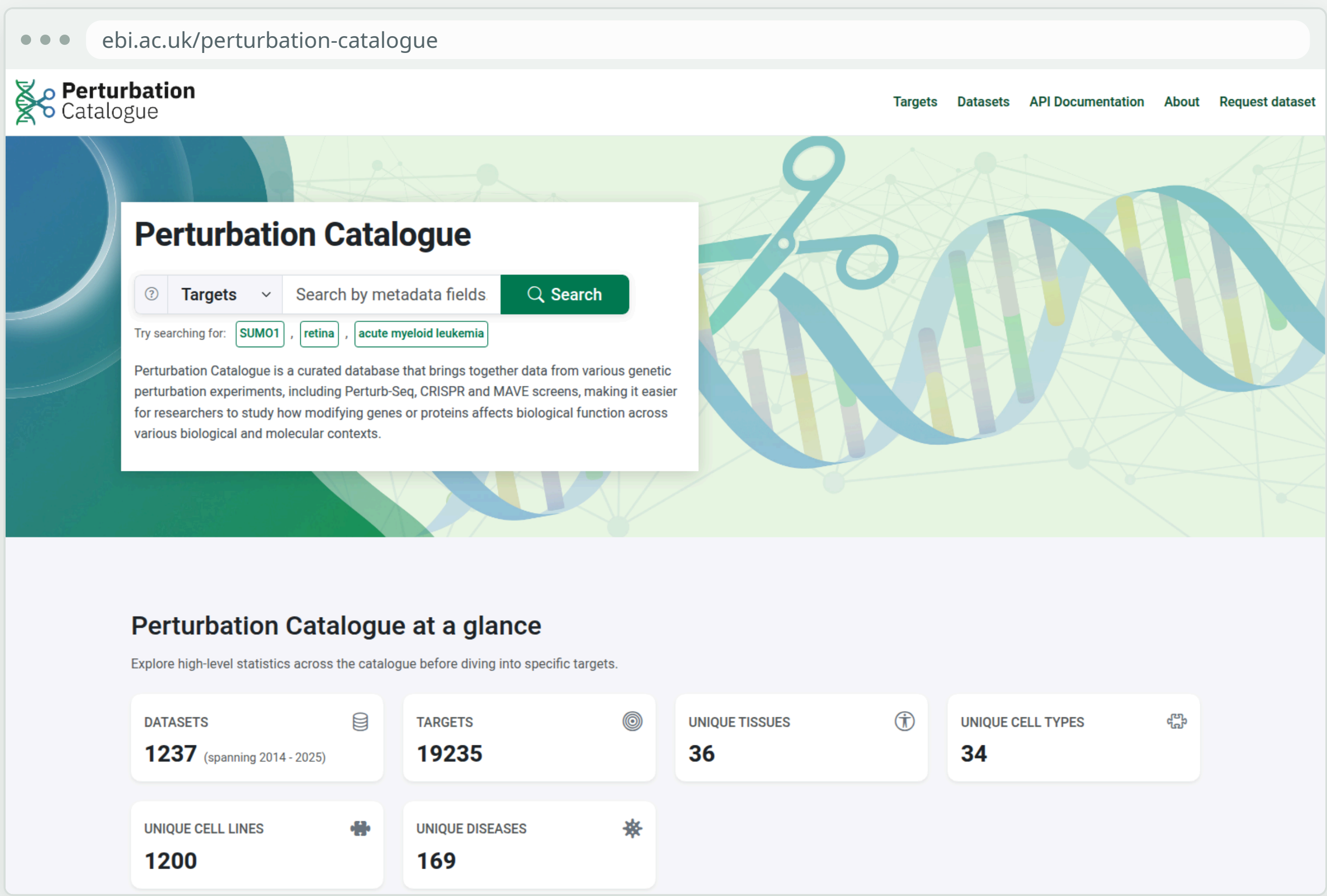
→

QC + filtering

Produce the final analysis-ready H5AD

Five datasets are live. Perturbation–control comparisons feed DEA and Hallmark GSEA. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Browse the Catalogue



Target search, current coverage and biological-context summaries in one human-facing portal.

Access records via API

Search and retrieve metadata or modality results through the same backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
```

```
{
  "perturbation": {"gene_symbol": "TFAM"},
  "effect": {"gene_symbol": "MT-ND5", "log2fc": -1.926, "padj": 0.0}
```

Download whole datasets

Take complete modality tables with provenance and metadata for local analysis or archival use.

Parquet

Columnar, efficient for large-scale analysis

CSV.gz

Portable, compressed tabular export

JSON metadata

Machine-readable dataset context and provenance

More exciting things are coming soon

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.